



United Energy Tariff Report 2005

2005 Network Tariff Report

Produced pursuant to clause 11.3.5 of Volume I of the Office of the Regulator General's Electricity Distribution Price Determination 2001- 2005, and supplementary advisory notes

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1 Introduction and Purpose

This report is intended to give guidance to customers, setting out the issues, principles and rationale used by United Energy Distribution (UED) in setting its Network Use of System (NUOS) tariffs for 2005, the final year of the 2001-2005 regulatory period. However, UED is currently in the process of re-setting its network tariffs for the next regulatory period 2006-2010 and has submitted a detailed price-service proposal which presents important background information on how tariffs have changed over the current regulatory period as well as UED's proposals for future developments. It also provides detailed information on future service level benchmarks and proposed reductions in distribution charges. It is recommended that this 2005 tariff report be read in conjunction with the price-service proposal as well as the previous four tariff reports, available on the Essential Services Commission (Commission) (www.esc.vic.gov.au) and UED's (www.unitedenergy.com.au) websites.

UED believes that it is important to provide customers with reasonable information on the objectives and rationale for tariff design, and annual changes in tariffs. Prior to the commencement of the 2006-2010 regulatory period, UED will be publishing an overarching Tariff Strategy Report covering the 5 years from 2006, which will:

- set out their tariff strategy for that regulatory period (including details on the tariffs that the distributors are proposing to charge, as well as a list of excluded services charges and metering charges);
- provide a discussion of their tariff setting policy framework and an explanation of how they have had regard to the Commission's pricing principles;
- explain the basis upon which tariffs were developed including the methodology adopted for developing the structure and level of tariffs, and a description of the methodologies used to determine stand alone and avoidable costs;
- explain the basis on which they have determined the appropriate breakdown between fixed and variable charges;
- indicate the extent to which the tariff structures provide efficient consumption signals to distribution customers;
- indicate the methodology adopted for allocating transmission-related costs to distribution customers through their transmission tariffs;
- provide an estimate of the average annual distribution and transmission charge (in \$) for each combination of distribution and transmission tariff; and
- present at least the initial tariffs and a description of how the tariffs and tariff structure are likely to be adjusted over the period including, where new tariffs are to be introduced or old tariffs are to be made redundant.

The production of the Tariff Strategy Report will provide customers with substantial information regarding the future direction of tariffs and a detailed exposition of each distributor's methodology for setting tariffs.

Consultations with customers and groups representing customers have continued over 2004 and will continue into the future. However, UED welcomes further comments from interested parties and has demonstrated its willingness to incorporate such views into future tariff settings through some of the changes implemented over the past four years.

Updated: 4 August 2005

2 Tariff Setting Principles (Basis for Tariff Policies)

The design of the tariffs sought to incorporate the following policy principles:

- **Regulatory compliance.** Compliance with the regulatory requirements of the Office of the Regulator General (ORG) and the Commission, as specified in and interpreted from Volumes I & II of the Electricity Distribution Price Determination, and subsequent partial re-determination. A key requirement by the Commission is that the network prices faced by a group of customers must lie between the avoidable cost of servicing that group (lower bound) and the stand-alone costs of servicing that class (upper bound). These regulatory requirements limit the extent to which tariffs can change, particularly any average price increases.
- **Customer choice.** To provide customers with meaningful choices of tariff options based upon behavioural response and shared benefits.
- **Market equity.** Having regard to previous price levels and hence relative changes. Further, pricing should apply to all retailers in a neutral manner and not impede the viability of full retail contestability.
- **Cost reflectivity.** Seek appropriate mechanisms for achieving inter-customer group equity in the allocation of distribution cost-recovery requirements. Pricing to recognise cost-reflectivity within the constraints imposed by regulatory requirements (ie. that tariffs fall within economic upper and lower bounds) and ensuring efficient investment signalling and network usage.
- **Behavioural elasticity.** Seek to utilise rational consumer behavioural elasticity, both in terms of usage responses to pricing signals and tariff switching.
- **Practicality.** Seek to simplify mechanisms balancing against economic functionality, while having regard to minimising transaction and pricing administration cost.
- **Environmental.** Within the limitations of the scope and context of electricity distribution pricing, have regard to opportunities to improve utilisation efficiency and accommodate emerging energy technologies, particularly in respect of greenhouse gas emission reduction.

Understandably, not all the principles could be satisfied to their full extent, while fully complying with the Commission's requirements, particularly as some pull in opposing directions. UED has sought to satisfy these principles to the greatest extent possible whilst maintaining the given regulatory constraints.

3 Changes in Tariffs, Tariff Structure and New Tariffs

The 2005 Network Use of System (NUOS) tariffs reflect the underlying structure of both the Transmission Use of System (TUOS) and Distribution Use of System (DUOS) charges. That is, the structures of the TUOS and DUOS tariffs are identical and the NUOS rates are the simple addition of DUOS and TUOS.

The fundamental restructuring of UED's tariffs was undertaken with the 2001 tariff set. The 2002 tariff submission contained some minor restructuring including the replacement of the interruptible tariff introduced in 2001 and two new tariffs for small customers. The 2005 tariff set, as per the 2003 and 2004 tariff set contained no new tariffs.

4 Uptake of Time of Use (TOU) Tariffs

In 2001 and 2002, UED introduced a number of optional Time of Use (TOU) tariffs for customers for use in conjunction with interval (or smart) meters. At the same time, several tariffs incorporating the historical 11pm to 7am Off Peak time zone, relying on obsolete metering technology, were closed to new connections. The following tariffs are closed to new connections and only available to installations currently taking supply under this tariff:

1. Dedicated circuit
2. Low voltage large 1 rate
3. Low voltage large 2 rate
4. Low voltage medium 2 rate 5 day
5. Low voltage medium 2 rate 7 day
6. Low voltage small 2 rate
7. Subtransmission KVA time of use

The tariffs that are open to new connections are as follows:

1. High voltage KVA time of use
2. High voltage KVA time of use - HOT
3. Low voltage large KVA time of use
4. Low voltage large KVA time of use - HOT
5. Low voltage medium 1 rate
6. Reverse cycle airconditioning time of use
7. Low voltage small 1 rate
8. Winter economy tariff
9. Unmetered supplies

Further information on each of the above tariffs is available on UED's website: www.unitedenergy.com.au.

New connections requiring off peak network tariffs are now required to install an interval meter. It was intended that such a change in connection characteristics would trigger a change in the underlying network tariff to be billed to the customer's Retailer.

Previously, delays with the Commission's formal position paper on the roll out of interval metering, have meant that UED's uptake of TOU tariffs has been substantially lower than forecast at the time of introduction of the 2001 tariffs. A further related issue is the reluctance of Retailers to repackage UED's TOU tariffs to smaller customers. To date, no Retailer has publicly made an offer based on the low volume TOU tariff. Informal discussions have revealed a number of operational reasons associated with these decisions, including that most customers currently would need to pay additional fees to have an interval meter installed to take advantage of the new tariffs.

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In July 2004, the Commission's announced a final decision to mandate a rollout of interval meters to all Victorian electricity customers. The Commission's final decision is to require all large customers to have interval meters by 2008 and to mandate new and replacement interval meters from 2006.

Interval meters provide the opportunity to measure the real time consumption of electricity in predetermined time periods and to use the information obtained to set end-use pricing to mirror wholesale market price movements, and thus more accurately reflect the real cost of supply. To date interval metering has mainly been used for customers consuming greater than 160 MWh per year and for those customers who have chosen to enter the contestable retail market with a contestable meter provider.

The Commission's decision to implement an interval metering strategy as proposed for the Victorian market is considered to be at the forefront of industry practice, not only in Australia but internationally, particularly in terms of the volume of metering installations and data quantities involved. As such it presents particular risks and uncertainties for Distributors. In particular, whereas UED manages the risks involved with its normal business operations, UED has no choice but to supply interval meters in accordance with the Commission's timetable. This creates risks that are beyond UED's control.

UED considers that the Commission's consultation process to date in relation to the mandated interval meter rollout, has not allowed a complete examination of these commercial issues associated with the rollout to occur. UED looks forward to discussing these issues with the Commission at the earliest possible opportunity. If the Commission considers its obligations under the rules of procedural fairness require it to deal with the issue through an open consultation process UED welcomes such transparency.

In 2001, UED introduced a number of tariff initiatives, which are now expected to have a greater influence on customer behaviour as interval meters are rolled out. UED will continue to review the effectiveness of its existing tariffs in the light of developments in metering technology. The rollout of interval metering represents an important development in the next regulatory period (2006-2010) that is likely to drive future tariff design. UED is committed to the progressive introduction of interval meters across all customer segments in the medium term, and will assess the tariff design implications as more accurate half-hourly load shapes for all customers becomes available. Against this background, UED is not proposing to mandate re-assignment to time-of-use tariffs in association with the rollout of interval meters. UED will maintain a 'watching brief' on the need for more substantive tariff changes and will assess the existing time-of-use tariffs as more accurate half-hourly load shapes for all customers becomes available. This would allow sufficient time for UED to consult with customers on the implications of the change and understanding of the demand component of the time-of-use tariff. Customers will then be able to make a more informed decision to take up the time-of-use tariffs.

5 Distribution Use of System (DUOS) Tariffs

The regulated price control formula allowed UED an average increase in the price control for distribution tariffs in 2005 of 1.616% over the 2004 tariffs. The major components of this increase were “CPI – X” (1.322%), and “S” (0.3129%). The term “CPI-X” is the underlying price path for distribution tariffs over the regulatory period (2001-2005). The “S” factor relates to network reliability and reflects the network reliability statistics achieved by UED’s management of its network assets.

Within the average tariff increase, the price control formulae allows individual tariffs to increase by a maximum of 4.695%, although the average increase across all customer classes is limited to 0.165%. This allows UED some flexibility as to where the average 0.165% DUOS price control increase is incurred.

All closed tariffs were increased by the maximum allowable price increase of 4.695%, which allowed the open tariffs (notably the TOU tariffs) to receive a lower price increase of between 1.4% and 1.6% in DUOS price to meet the average 1.616% overall price increase. As previously explained, UED’s TOU tariff uptake has been below expectations, and this effective DUOS price decrease is an added incentive and reward to those customers who wish to take up a TOU tariff. Additionally, the LV large KVA tariff received a DUOS price reduction to make it more cost reflective as compared to the HV tariff.

Closed tariffs (non TOU) were increased by the maximum allowable to reflect their increasing contribution to co-incident peak demand due to the growth in cooling load amongst small and medium customers and the inability to distinguish between “dumb” profiles. This treats them all as if cooling load exists at the premise and customers in these classes without cooling load or who are prepared to respond to TOU price signals are incentivised to switch to one of the optional TOU tariffs available.

As mentioned in Section 2, the Commission has stated that distribution tariffs should lie between the following upper and lower bounds:

- tariffs for each customer should generate revenue in excess of the avoidable cost to service the customer; and
- tariffs for each customer should generate revenue less than the cost of providing the service on a stand-alone basis to the customer.

The Commission has not provided any further guidance on how the upper and lower bounds should be assessed. In UED’s view, there are a myriad of alternative methodologies that could be applied for estimating upper and lower bounds for network pricing and also for cost allocation to customer groups. Each methodology has its merits and shortcomings.

To demonstrate that distribution tariffs fall between the avoidable cost “floor” and standalone cost “ceiling”, UED must first apply a “cost of supply” methodology to assist in setting tariff rates. Broadly speaking, tariff rates are set to recover the allocated distribution revenue from that customer group. It is noted, however, that UED has regard to all the pricing principles outlined in Section 2, in setting tariff rates. Furthermore, in terms of tariff structure, UED has listened to its customers, who have expressed a strong preference for reductions in fixed charges.

The critical issue from a cost of supply modelling perspective is the method by which distribution revenue is allocated across the tariff groups. As network businesses are characterised by relatively high fixed costs and significant asset-sharing between

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customer groups, there is no unambiguously “correct” method for allocating costs. UED’s method of allocation is based on each tariff’s relative usage of UED’s network assets.

The “cost of supply” methodology and data inputs are described briefly below:

5.1 Load Profile Data

- Average load patterns for each tariff are derived from historical data such as total or sample tariff interval meter data, SCADA, zone sub, or terminal station interval data.
- Load patterns for each tariff are compared to the system profile, at each asset level. From this data, each tariff’s contribution to the maximum demand on the various asset levels of the network is determined.
- Load patterns are also used to determine average tariff parameters such as load factor and power factor.
- In the case of unmetered supplies, a load pattern is assessed from installed equipment inventory, wattage ratings and daylight hours.

5.2 Asset Data

- UED’s network asset database is used to determine the number of asset “levels” and value of the assets employed to serve each tariff.
- The asset “levels” attributed to each tariff is determined in accordance with the following ‘asset participation matrix’.

Table 5:1 Asset Participation Matrix

Tariff name	Asset level				
	SubTrans	High V	Low Voltage (LV)		
	SubT	High V	Large	Medium	Small
Unmetered supplies					
Low voltage small 1 rate					
Low voltage small 2 rate*					
Dedicated circuit*					
Winter economy tariff					
Low voltage large KVA time of use					
Low voltage medium 1 rate					
Low voltage medium 2 rate 5 day*					
Low voltage medium 2 rate 7 day*					
Reverse cycle airconditioning time of use					
Low voltage large 2 rate*					
Low voltage large KW time of use					
Low voltage large KW time of use - HOT					
Low voltage large 1 rate*					
Low voltage large KVA time of use					
Low voltage large KVA time of use - HOT					
High voltage KVA time of use					
High voltage KVA time of use - HOT					
Subtransmission KVA time of use*					
LVS1R					LV Small class
LVS2R					
LVDed					
WET					
LVKWTOU (sml:med)				LVM(cont)	LVS(cont)
LVM1R					
LVM2R5D					
LVM2R7D					
RCACTOU					
LVL2R					
LVLKWTOU					
LVLKWTOU Interruptable (lge)					
LVL1R					
LVKVATOU (lge)					
LVKVATOU Interruptable (lge)					
HV-KVA					
HV-KVA Interruptable					
ST22.KVA					

* Tariff closed to new connections and premises not already taking supply under this tariff.

UED has extended its “cost of supply” methodology to assess the avoidable and standalone costs. The avoidable cost model recognises that only a proportion of total costs are avoidable. In particular, the majority of asset-related costs cannot be avoided even if a particular customer group is no longer served. Inevitably, the assessment of which costs are avoidable is a matter of judgement. It should be noted, however, that as the avoidable costs are less than the total costs, UED’s cost of supply methodology will always set tariffs at a level that exceeds avoidable costs.

UED’s modelling of standalone costs is similarly based on the cost of supply model. The principal differences between the “basic” cost of supply estimates and standalone costs are:

- standalone networks to serve a particular tariff class will not enjoy the benefit of diversity in peak demand between tariff classes;
- economies of scale may be lost in supplying a subset of existing customers or tariffs;
- greater urban congestion may result in the optimised replacement cost exceeding UED’s regulated asset value; and
- it is likely that a notional “standalone” competitor to UED may seek a rate of return that exceeds the regulated cost of capital.

These factors indicate that the standalone costs will exceed the cost of supply estimates on which UED bases its tariff design. It is important to recognise that it is difficult to determine the standalone costs with precision – inevitably a judgement justbe made. The results of UED’s modelling is summarised in Table 5.2 below:

Table 5.2: Comparison of 2005 Proposed Tariff Rates with Estimated “Cost Window”

Tariff Group		Lower Bound "Avoidable Cost" (c/kWh)	2005 Avg DUOS (Exc GST) (c/kWh)	Upper Bound "Standalone Cost" (c/kWh)
LV Unmetered		0.16	3.53	6.97
LVS1R	Small	0.28	4.97	10.41
LVS2R*			4.04	
LVDed*			1.41	
WET			5.32	
LVkWTOU			1.84	
LVM1R	Med	0.35	7.57	12.44
LVM2R5D*			6.36	
LVM2R7D*			9.02	
LVL2R*	Large	0.35	5.42	12.44
LVL1R*			4.77	
LVKVATOU (lge)		0.12	2.84	4.28
HV-KVA	HV	0.06	1.61	2.12
HV-KVA.Interruptable			**	
ST22.KVA*			0.40	

* Tariff closed to new connections and customers not already taking supply under this tariff

** Omitted due to small number of customers in this tariff class

6 Transmission Use of System (TUOS) Tariffs

For practical reasons, the TUOS tariff structure does not reflect the structure of the grid fees paid. The TUOS tariff commenced with a structure compatible with the DUOS tariff. The structure has been maintained in order to allow the NUOS tariff to be determined by simply adding the DUOS and TUOS rates. UED has however restricted the application of TUOS rates to those components of the NUOS which best reflect the underlying Grid Fees, (that is Peak Energy, Summer Demand Incentive Charge and Rolling Demand) and so Off Peak energy and fixed charges do not attract TUOS.

In 2004, the Commission restricted the average TUOS price rise to 23.1% when it needed to be 49.6% in order to recover costs. This year, the Commission has again restricted the average TUOS price increase. In order to recover costs, UED's 2005 average TUOS rate needed to increase by 75.3%. However, the Commission has restricted the increase to 23%. The increase of 75.3% is primarily driven by two significant factors – these being:

6.1 Annual Adjustment

Once again the Commission imposed a 20% cap on prior year grid fees to be recovered by UED. This decision means that UED carries a further cashflow cost of \$13M in unfunded grid fees for 2004. This exposes UED to significant regulatory risk in terms of cashflow and distorts TUOS passthrough pricing signals to the market.

6.2 Land Tax

The Commission has previously recognised that the existing price controls have not allowed UED to pass through costs incurred for land tax for the period 1 July 2004 – 31 December 2004. This was considered appropriate by the Commission on the basis that the Victorian Government's removal of the smelter reduction levy on retail business, effective 1 July 2004, should result in no net impact on customers. In addition to this, the introduction of land tax on transmission easements results in an increase in TUOS to UED compared to the previous charges. The component relating to land tax requires an increase in the Y Factor of 41.4%.

6.2.1 *Issues with the methodology for Transmission cost pass through - Land tax*

As discussed in the Commission Open Letter, "Final Decision to increase the 2005 Rebalancing Control for Transmission Tariffs" on 12 November 2004, the Commission's usual approach "to Y factor variation would prevent land tax pass through in a manner that replicates the previous customer impact of the smelter reduction levy". "Some large customers would face price increases greater than if the land tax cost pass through was delivered via the same mechanism as the former smelter reduction levy. That is, via a cents per kilowatt hour (c/kWh) charge on energy consumption".

On 20 October, the Commission introduced a second methodology adopting a two part approach to Y factor variation in 2005.

This approach required the adjustment to account for other pass through items of up to 20 per cent, be applied first, with the application of a specific c/kWh addition to each tariff for land tax pass through. With regards to the pass through of the land tax, it involves the calculation of revenue required to achieve the pass through in a c/kWh

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form for each tariff, then calculation of the required year on year movement in revenue to obtain a percentage variation for each tariff.

The Commission found the application of a specific Y factor rebalancing control for each transmission tariff is consistent with the current price controls and has approved this approach for a specific Y factor variation for each tariff of each distribution business. Individual 2005 Y factor variations for UED tariffs are provided in table 6.1 below.

Table 6:1 2005 Y Factor

United Energy		
Tariff	Land tax impact on Y factor	2005 Y factor
Low voltage small 1 rate	29.3%	49.3%
Winter economy tariff	29.9%	49.9%
Low voltage small 2 rate	57.5%	77.5%
Low voltage medium 1 rate	25.2%	45.2%
Low voltage medium 2 rate 5 day	66.0%	86.0%
Low voltage medium 2 rate 7 day	34.5%	54.5%
Unmetered supplies	90.7%	110.7%
Low voltage large 1 rate	25.1%	45.1%
Low voltage large 2 rate	65.3%	85.3%
Low voltage large KW time of use	50.0%	70.0%
Low voltage large KVA time of use	53.3%	73.3%
High voltage KVA time of use	78.9%	98.9%
Subtransmission KVA time of use	76.2%	96.2%

In its open letter, the Commission concluded that by adopting this approach, it enables (1) the increased TUoS costs associated with the introduction of land tax to be passed through in a manner that maintains the same proportional cost shares previously incurred by customers under the smelter levy and (2) maintains consistency with the current Transmission tariff rebalancing control formula in Clause 3.4.2 of the Volume II - Price Controls”.

The Commission’s open letter can be obtained from the Commission’s website: <http://www.esc.vic.gov.au/electricity139.html>.

7 Network Use of System (NUOS) Tariffs

NUOS charges are those seen on customer invoices and reflect the underlying structure of Transmission and Distribution charges. Residential and small business customers do not see distribution prices explicitly, as they are usually bundled by retailers with energy and other costs. NUOS rates are the simple addition of DUOS and TUOS rates. As a result of the movements discussed above, the NUOS rates are as follows:

Schedule of Network Use of System (NUOS) Tariffs: 1 January 2005 (GST inclusive)

Tariff	Component							Eligibility		Min.
	Fxd	Egy SmrPk	EgyNsm Pkb1	EgyNsm Pkb2	Egy Opk	Dmn Rln	Dmn Smr	kVA	MWh	DmnRln
Dedicated circuit*					1.758					
High voltage KVA time of use		1.474	1.158		0.658	13.304	15.696			1,150
High voltage KVA time of use - HOT		1.109	0.865		0.658	10.874	25.345			1,150
Low voltage large KW time of use		8.228	6.377		1.506		38.283	<150	<400	
Low voltage large KW time of use - HOT		7.585	5.862		1.506		66.900	<150	<400	
Low voltage large 1 rate*	11.531	8.209	6.533							
Low voltage large 2 rate*	16.623	11.938	9.333		1.758					
Low voltage large KVA time of use		2.040	1.603		0.925	16.030	20.187	>150	>400	150
Low voltage large KVA time of use - HOT		1.617	1.263		0.925	14.299	34.371	>150	>400	150
Low voltage medium 1 rate	10.627	10.544	8.444					Business Customer		
Low voltage medium 2 rate 5 day*	16.623	13.408	10.501		1.758					
Low voltage medium 2 rate 7 day*	16.623	13.420	10.563		1.758					
Reverse cycle airconditioning time of use		18.831	4.343		1.621		75.058	<20	<70	
Low voltage small 1 rate	8.175	7.909	6.134					Residential Customer		
Winter economy tariff	8.175	7.918	6.141	3.257				<20	<70	
Low voltage small 2 rate*	12.784	12.701	9.686		1.758					
Subtransmission KVA time of use*		0.895	0.697		0.300	4.166	5.234			11,100
Unmetered supplies		12.855	9.710		1.758					

* Tariff closed to new connections and premises not already taking supply under this tariff.

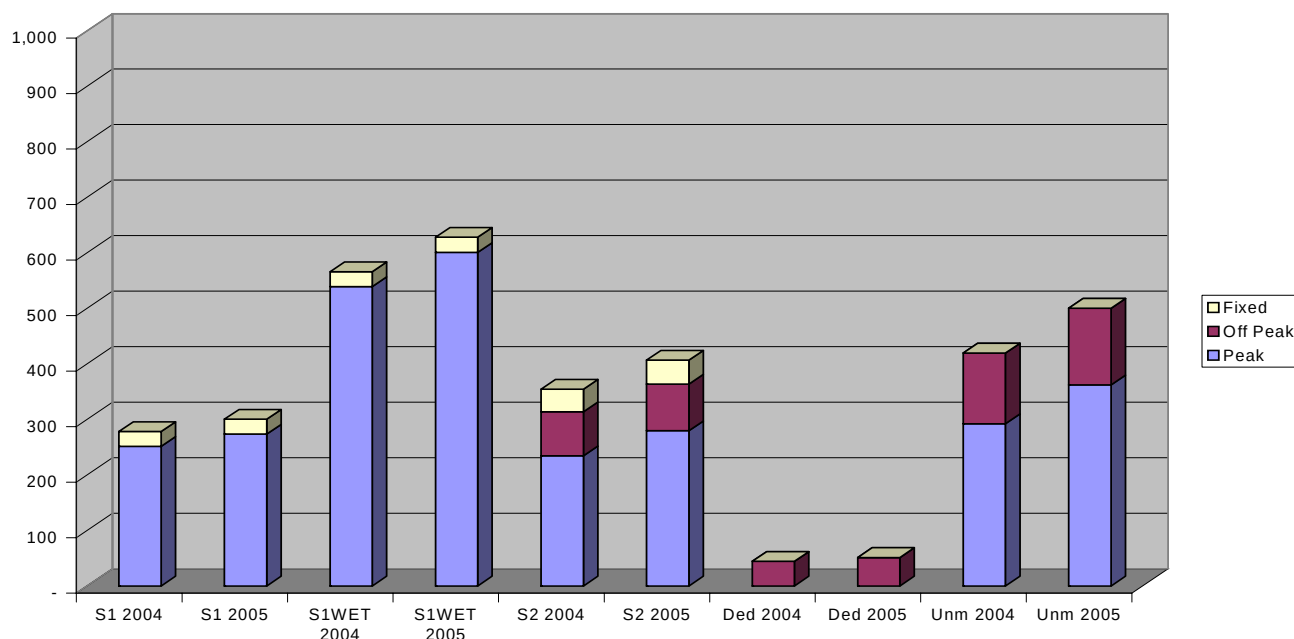
Components

Abbreviation	Description	Units
Fxd	Fixed charge	c/day
EgySmrPk	Energy - Summer peak or single rate	c/kWh
EgyNsmPkb1	Energy - Non-summer peak or single rate, block 1	c/kWh
EgyNsmPkb1	Energy - Non-summer peak or single rate, block 2	c/kWh
EgyOpk	Energy - Off-Peak	c/kWh
DmnRln	Demand - Rolling 12 months	c/kVA/day
DmnSmr	Demand - Summer Incentive	c/kW/day or c/kVA/day

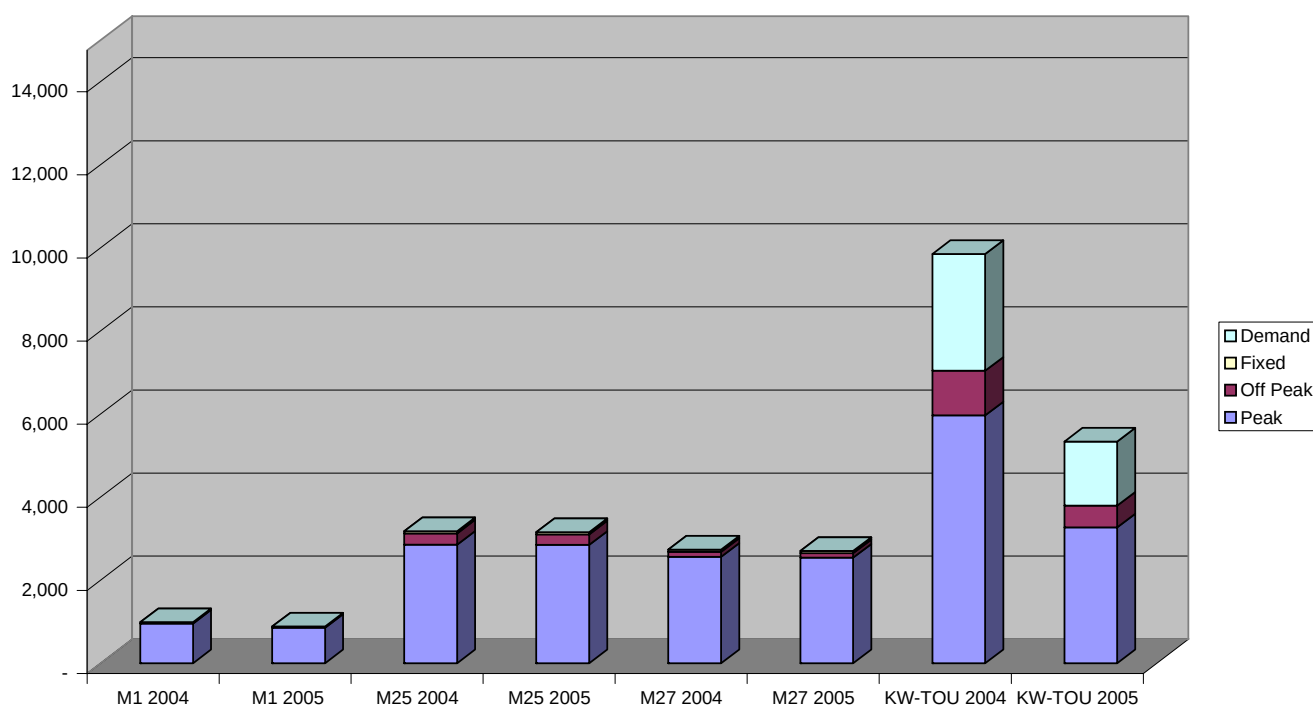
7.1 Average charge per customer

This section presents the average yearly charge that United Energy's customers received in 2004 and are estimated to receive in 2005. The following graphs are set out in the four main customer categories; small, medium, large and HV.

Average Distribution and Transmission charge per customer - Small

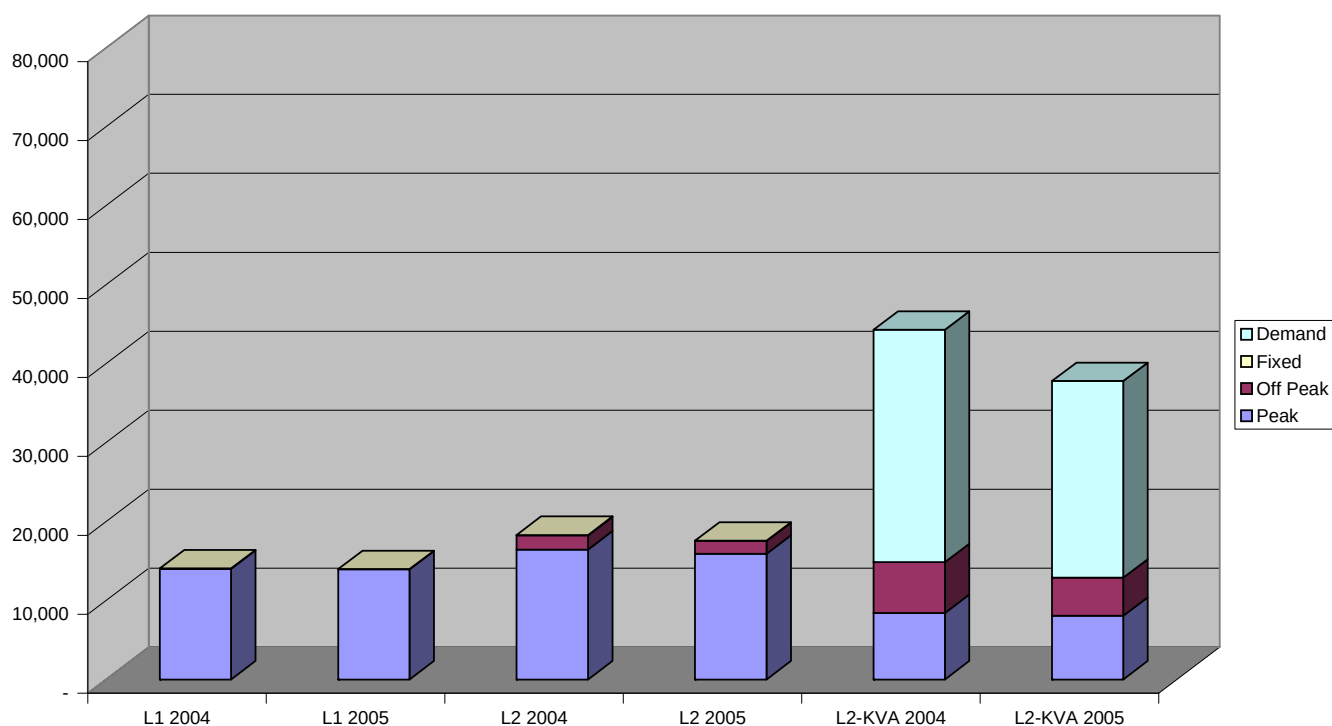


Average Distribution and Transmission charge per customer - Medium

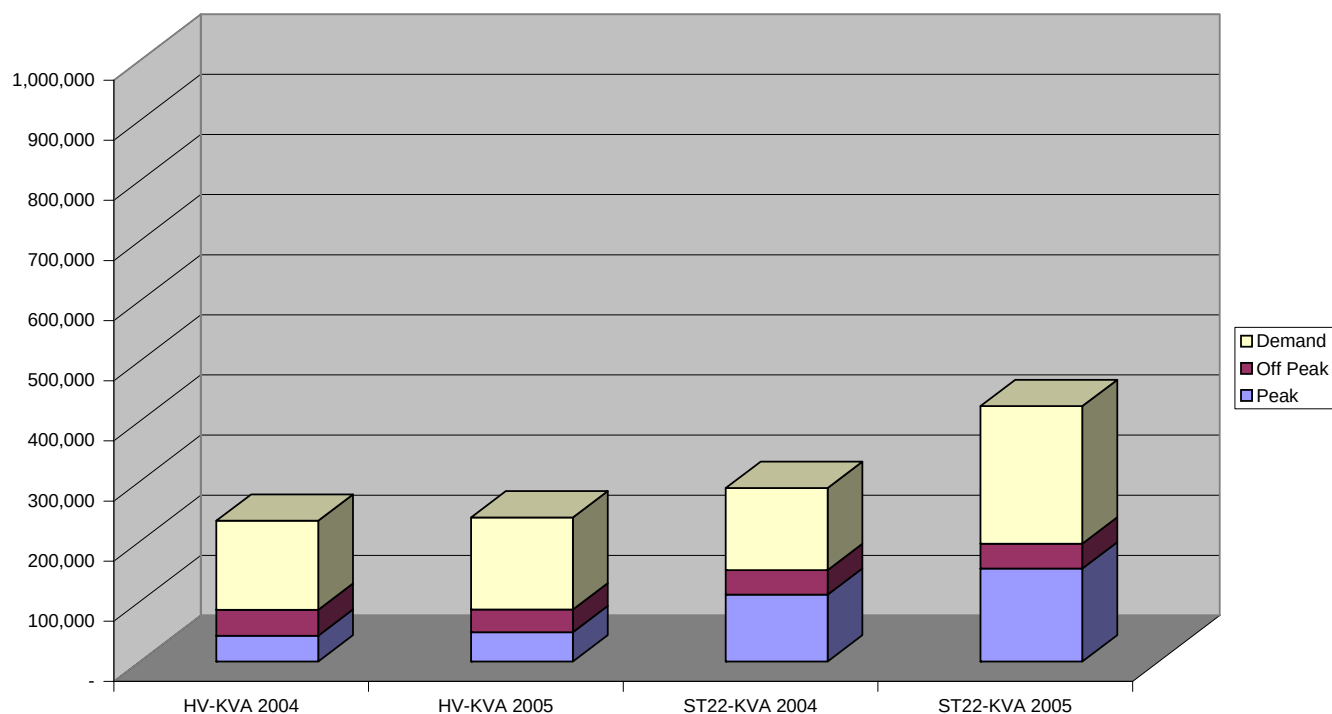


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Average Distribution and Transmission charge per customer - Large



Average Distribution and Transmission charge per customer - HV



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8 Future Direction of UED's Tariffs

UED will continue to review the effectiveness of its existing tariffs in the light of developments in metering technology. The rollout of interval metering represents an important development in the next regulatory period that is likely to drive future tariff design.

As noted in Section 13.3 of UED's 2006 Price Services Offering UED has been a leader in tariff innovation and is committed to the progressive introduction of interval meters across all customer segments in the medium term, and will assess the tariff design implications as more accurate half-hourly load shapes for all customers becomes available.

It is expected that while there may be some refinements to existing tariffs, at this stage, UED is not contemplating significant tariff changes during the 2006-2010 regulatory period. That said, UED will maintain a watching brief on the need for more substantive tariff changes.

- Encouraging an increased uptake of interval-metering based tariffs such as the current 'TOU' series;
- The SDIC concept will remain, but the time window may be updated from time to time in order that it remain aligned with the key network peak demands;
- Cost-of-supply modelling updated to reflect changes in relative contributions from segments;
- Further accentuating the emphasis on peak season (summer), day of week and time of day in order to stimulate DSM response;
- Properly integrate the contribution that distribution-connected generator customers should be making to the costs of providing network services that all users share and derive value from;
- Further closure of tariffs based on obsolete metering;
- Premium service tariffs whereby customers get a choice of above code-level supply reliability and services, for a premium on top of the standard tariff. This must be seen in the overall context of customer service as well as relationship strategies;
- An increased number of time-of-day bands, with greater peak / off peak differential, and energy and distribution tariff components peaking at different times; and
- Demand management (DM) programs aimed at different customer classes may be investigated, for example:
 - interruptible tariffs for business customers whereby customers agree to reduce their power consumption for agreed periods at the request of the distributor (likely to be at a time like a hot summer afternoon when the time when the system is heavily stressed), and in return get some compensation payments from the distributor; and
 - DM aggregation program, which involves working with a range of customers and bidding their combined interruptible load in either the wholesale energy or ancillary services market.

9 Let us know your views

UED continues to encourage customers to discuss ideas on how mutual needs may be better met. Many of the 2005 changes will be familiar to those customers that participated in sessions hosted by UE during 2004. It is hoped that this can be further enhanced during the next year.

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Appendix A: 2005 NUOS, DUOS and TUOS Tariff Rates

Available on the Commission's website at:

<http://www.esc.vic.gov.au/electricity210.html>

and the UED website at:

http://www.unitedenergy.com.au/industry/network_tariffs/download/2005GSTExclusiveTariffs.pdf

DUOS Tariffs (GST inclusive) for 2005

Tariff	Component							Eligibility		Min.
	Fxd	Egy SmrPk	EgyNsm Pkb1	EgyNsm Pkb2	Egy Opk	Dmn Rln	Dmn Smr	kVA	MWh	
Dedicated circuit*					1.758					
High voltage KVA time of use		0.739	0.569		0.658	8.417	10.563			1,150
High voltage KVA time of use - HOT		0.739	0.569		0.658	8.417	20.383			1,150
Low voltage large KW time of use		6.664	5.125		1.506		30.989	<150	<400	
Low voltage large KW time of use - HOT		6.664	5.125		1.506		58.649	<150	<400	
Low voltage large 1 rate*	11.531	6.016	4.778							
Low voltage large 2 rate*	16.623	10.204	7.944		1.758					
Low voltage large KVA time of use		1.038	0.799		0.925	11.936	14.592	>150	>400	150
Low voltage large KVA time of use - HOT		1.038	0.799		0.925	11.936	28.160	>150	>400	150
Low voltage medium 1 rate	10.627	8.348	6.687					Business Customer		
Low voltage medium 2 rate 5 day*	16.623	11.667	9.106		1.758					
Low voltage medium 2 rate 7 day*	16.623	11.206	8.790		1.758					
Reverse cycle airconditioning time of use		5.290	1.330		1.621		66.472	<20	<70	
Low voltage small 1 rate	8.175	5.958	4.573					Residential Customer		
Winter economy tariff	8.175	5.958	4.573	1.689				<20	<70	
Low voltage small 2 rate*	12.784	9.423	7.060		1.758					
Subtransmission KVA time of use*		0.408	0.304		0.300	0.714	1.046			11,100
Unmetered supplies		9.668	7.161		1.758					

* Tariff closed to new connections and premises not already taking supply under this tariff.

Components

Abbreviation	Description	Units
Fxd	Fixed charge	c/day
EgySmrPk	Energy - Summer peak or single rate	c/kWh
EgyNsmPkb1	Energy - Non-summer peak or single rate, block 1	c/kWh
EgyNsmPkb1	Energy - Non-summer peak or single rate, block 2	c/kWh
EgyOpk	Energy - Off-Peak	c/kWh
DmnRln	Demand - Rolling 12 months	c/kVA/day
DmnSmr	Demand - Summer Incentive	c/kW/day or c/kVA/day

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TUOS Tariffs (GST inclusive) for 2005

Tariff	Component							Eligibility		Min.
	Fxd	EgySmrPk	EgyNsmPkb1	EgyNsmPkb2	EgyOpk	DmnRln	DmnSmr	kVA	MWh	
Dedicated circuit*										
High voltage KVA time of use		0.735	0.589			4.887	5.133			1,150
High voltage KVA time of use - HOT		0.370	0.296			2.457	4.962			1,150
Low voltage large KW time of use		1.564	1.252				7.294	<150	<400	
Low voltage large KW time of use - HOT		0.921	0.737				8.251	<150	<400	
Low voltage large 1 rate*		2.193	1.755							
Low voltage large 2 rate*		1.734	1.389							
Low voltage large KVA time of use		1.002	0.804			4.094	5.595	>150	>400	150
Low voltage large KVA time of use - HOT		0.579	0.464			2.363	6.211	>150	>400	150
Low voltage medium 1 rate		2.196	1.757					Business Customer		
Low voltage medium 2 rate 5 day*		1.741	1.395							
Low voltage medium 2 rate 7 day*		2.214	1.773							
Reverse cycle airconditioning time of use		13.541	3.013				8.586	<20	<70	
Low voltage small 1 rate		1.951	1.561					Residential Customer		
Winter economy tariff		1.960	1.568	1.568				<20	<70	
Low voltage small 2 rate*		3.278	2.626							
Subtransmission KVA time of use*		0.487	0.393			3.452	4.188			11,100
Unmetered supplies		3.187	2.549							
* Tariff closed to new connections and premises not already taking supply under this tariff.										

Components

Abbreviation	Description	Units
Fxd	Fixed charge	c/day
EgySmrPk	Energy - Summer peak or single rate	c/kWh
EgyNsmPkb1	Energy - Non-summer peak or single rate, block 1	c/kWh
EgyNsmPkb2	Energy - Non-summer peak or single rate, block 2	c/kWh
EgyOpk	Energy - Off-Peak	c/kWh
DmnRln	Demand - Rolling 12 months	c/kVA/day
DmnSmr	Demand - Summer Incentive	c/kW/day or c/kVA/day

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NUOS Tariffs (GST inclusive) for 2005

Tariff	Component							Eligibility		Min.
	Fxd	Egy SmrPk	EgyNsm Pkb1	EgyNsm Pkb2	Egy Opk	Dmn Rln	Dmn Smr	kVA	MWh	DmnRln
Dedicated circuit*					1.758					
High voltage KVA time of use		1.474	1.158		0.658	13.304	15.696			1,150
High voltage KVA time of use - HOT		1.109	0.865		0.658	10.874	25.345			1,150
Low voltage large KW time of use		8.228	6.377		1.506		38.283	<150	<400	
Low voltage large KW time of use - HOT		7.585	5.862		1.506		66.900	<150	<400	
Low voltage large 1 rate*	11.531	8.209	6.533							
Low voltage large 2 rate*	16.623	11.938	9.333		1.758					
Low voltage large KVA time of use		2.040	1.603		0.925	16.030	20.187	>150	>400	150
Low voltage large KVA time of use - HOT		1.617	1.263		0.925	14.299	34.371	>150	>400	150
Low voltage medium 1 rate	10.627	10.544	8.444					Business Customer		
Low voltage medium 2 rate 5 day*	16.623	13.408	10.501		1.758					
Low voltage medium 2 rate 7 day*	16.623	13.420	10.563		1.758					
Reverse cycle airconditioning time of use		18.831	4.343		1.621		75.058	<20	<70	
Low voltage small 1 rate	8.175	7.909	6.134					Residential Customer		
Winter economy tariff	8.175	7.918	6.141	3.257				<20	<70	
Low voltage small 2 rate*	12.784	12.701	9.686		1.758					
Subtransmission KVA time of use*		0.895	0.697		0.300	4.166	5.234			11,100
Unmetered supplies		12.855	9.710		1.758					

* Tariff closed to new connections and premises not already taking supply under this tariff.

Components

Abbreviation	Description	Units
Fxd	Fixed charge	c/day
EgySmrPk	Energy - Summer peak or single rate	c/kWh
EgyNsmPkb1	Energy - Non-summer peak or single rate, block 1	c/kWh
EgyNsmPkb2	Energy - Non-summer peak or single rate, block 2	c/kWh
EgyOpk	Energy - Off-Peak	c/kWh
DmnRln	Demand - Rolling 12 months	c/kVA/day
DmnSmr	Demand - Summer Incentive	c/kW/day or c/kVA/day

Appendix B: Glossary of Terms

DSM	Demand-side management. Refers to the ability of customers that consume a product to change behaviours that alter the overall balance between supply and demand. Examples may include the installation of energy management systems, or deferring consumption at time of peak demand.
Essential Service Commission	The Commission (Previously Office of the Regulator-General)
Load Duration Curve (LDC)	A graphical representation of the utilisation of a network. For the time period under observation, the load on the network is shown, sorted by highest to lowest. The resulting area graphically presents the percentage of the available capacity utilised.
Load Factor	Load factor is a measure of how 'steady' a load is. It is the extent to which average usage compares to maximum usage. It is generally much more efficient to supply a high load factor (100%=constant unchanging demand) than a 'peaky' (less than 100%) usage pattern. This is because depending upon the time of the peak, the capacity needs to be there all the time to supply the peak, even though most of the time it may not be fully required.
LVS1R etc	Abbreviated Tariff names. • The 1st two letters refer to the voltage level at which the customer is supplied, LV = low voltage 240V(single phase) or 415V(three phase), HV = high voltage 6.6, 11 or 22kV, ST = sub-transmission 22 and 66kV. HOT = tariff variant based on high ambient temperature threshold • The 2nd section refers to the type of customer, S = small, M = medium, L = large, no letters generally indicates 'generic' interval metered customer. • 3rd section indicates type of rate structure, 1R = single energy rate, 2R = peak/off-peak rate, kW = interval metered in kW, kVA = interval metered in kVA.
kW	Kilowatts, (1,000 Watts) the ISO unit of power measurement
kWh	An I.S.O. unit used for the measurement of energy. It is the product of kW times time (in hours).
kVA	kilo-Volt-Ampere, the vector product of voltage times current. It is a measure of the proportion of useful (kW) power delivered vs. the current (Amps) drawn by the load, this is known as 'power factor'. The size of assets required is generally a function of the current. Therefore the closer the power factor is to 1.00, the smaller the asset required to supply the load.

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k-factor mechanism	A regulatory method aimed at ensuring differences between 'actual' and 'forecasted' are washed-up year to year on a revenue neutral basis. This applies to Transmission wholesale costs and TUOS revenues.
Residue settlement	A means by which over-recoveries of costs on the part of the energy and transmission markets are rebated back to customers.
SDIC	Summer Demand Incentive Charge